

A Comparative Study of Numerical Methods for European Option Pricing

Black-Scholes · CRR Binomial Trees · Monte Carlo Simulation
with Variance Reduction · Heston Stochastic Volatility Model
· MLP Pricer · LSTM-BS Hybrid · VAE Implied Volatility Surface

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Abstract

This paper presents a systematic, reproducible comparison of classical numerical methods, stochastic volatility models, and modern deep learning approaches for pricing European options. Seven methodologies are implemented from scratch in Python and evaluated head-to-head on 200 live SPY call options (June 2026). We benchmark the Black-Scholes analytical formula, the Cox-Ross-Rubinstein binomial lattice, Monte Carlo simulation with antithetic and control-variate variance reduction, and the Heston (1993) stochastic volatility model — alongside three deep learning architectures: a Multi-Layer Perceptron (MLP), an LSTM-based volatility forecaster, and a Variational Autoencoder (VAE) for implied-volatility surface generation. Classical GBM methods cluster at MAE \$0.904 with near-identical accuracy; the Heston model achieves MAE \$1.04 while being the only method that structurally reproduces the empirical negative-skew implied-volatility smile ($\rho = -0.7$, IV range 17.3%–23.0%). Deep learning architectures do not outperform Black-Scholes on point accuracy for vanilla calls but

1. Introduction

Options pricing is a fundamental problem in quantitative finance. Black and Scholes (1973) established the benchmark for pricing European options under constant volatility and continuous trading. However, this foundational model has a well-documented limitation — the volatility smile: empirical markets consistently exhibit a negative skew (higher implied volatility for OTM puts), directly contradicting the log-normal GBM assumption.

Over five decades, researchers developed numerical and stochastic methods to address this. Cox, Ross, and Rubinstein (1979) introduced the binomial lattice, enabling American-option pricing via backward induction. Boyle (1977) pioneered Monte Carlo simulation for path-dependent exotics. Heston (1993) introduced a stochastic volatility model in which variance follows a mean-reverting CIR diffusion correlated with the spot — providing the structural ability to price options consistently across strikes.

More recently, machine learning has emerged as a powerful tool to bridge classical limits and empirical realities. Neural networks offer a promising avenue to learn complex, non-linear pricing manifolds from market data or to enhance existing models by forecasting dynamic parameters without rigid parametric assumptions. This paper presents a systematic, reproducible comparison of all these approaches.

Our key contributions: (1) Comprehensive implementation of seven distinct methodologies from scratch in Python. (2) Smile-consistent pricing analysis — quantifying why flat-volatility models fail and how Heston and VAE architectures reconstruct the empirical volatility surface. (3) Rigorous head-to-head benchmarking on 200 live SPY options across varying moneyness levels, offering a realistic speed-accuracy tradeoff assessment.

2. Related Work

The mathematical pricing of options has evolved through distinct phases. Black and Scholes (1973) derived the first closed-form solution for European options under constant volatility. Cox, Ross, and Rubinstein (1979) discretised GBM into a recombining lattice with early-exercise capability. Boyle (1977) pioneered Monte Carlo simulation. Heston (1993) introduced stochastic variance following a

3. Mathematical Foundations

3.1 Black-Scholes Model

The Black-Scholes (1973) framework prices a European call under risk-neutral GBM dynamics:

$$C = S N(d_1) - K e^{-rT} N(d_2)$$
$$d_1 = \frac{\ln(S/K) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

where $N(\cdot)$ is the standard normal CDF. Analytical Greeks follow by differentiation.

3.2 Cox-Ross-Rubinstein Binomial Tree

The CRR (1979) lattice calibrates up/down factors to match continuous-time GBM moments:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u}, \quad p = \frac{e^{r\Delta t} - d}{u - d}$$

Backward induction from terminal payoff $\max(S_N - K, 0)$ with early-exercise check at each node enables American-style pricing.

3.3 Monte Carlo Simulation with Variance Reduction

Under the risk-neutral measure, the terminal spot price under GBM is:

$$S_T = S_0 \exp \left[\left(r - \frac{1}{2}\sigma^2 \right) T + \sigma\sqrt{T} Z \right], \quad Z \sim \mathcal{N}(0, 1)$$

Variance reduction: (i) Antithetic Variates — pairing each path with its negated Brownian halves estimator variance. (ii) Control Variates — using the known expected asset price as a baseline correction, yielding 2.62x empirical variance reduction.

3.4 Heston Stochastic Volatility Model

Heston (1993) extends GBM by making instantaneous variance v_t itself stochastic:

$$dS = rS dt + \sqrt{v} S dW_1$$
$$dv = \kappa(\theta - v) dt + \sigma_v \sqrt{v} dW_2$$
$$\text{Corr}(dW_1, dW_2) = \rho dt, \quad \text{Feller: } 2\kappa\theta > \sigma_v^2$$

Negative ρ generates the empirical left skew: OTM puts carry higher implied vol than OTM calls. Parameters: $\kappa=2, \theta=0.04, \sigma_v=0.3, \rho=-0.7, v_0=0.04$.

4. Methodology — Model Descriptions

Black-Scholes (BS)

Closed-form analytical solution (Eq. 1). Five inputs: S , K , T , r , σ . Vectorised; execution time < 0.01 ms per option. Greeks computed analytically.

CRR Binomial Tree

$N = 200$ time steps, $O(N^2)$ backward induction. Per-node early-exercise check. Converges to Black-Scholes as $N \rightarrow \infty$.

Monte Carlo with Variance Reduction

100,000 paths with antithetic variates and control variates. Std error ≈ 0.057 at $N = 10,000$ paths. Flexible for exotic/path-dependent payoffs.

Heston MC

Euler-Maruyama discretisation, $N = 252$ daily steps, 50,000 antithetic paths (smile plot); 10,000 paths, $N = 63$ for evaluation speed. Full-truncation scheme for variance: $v_{t+\Delta t} = \max(v_t + \kappa(\theta - v_t)\Delta t + \sigma_v\sqrt{v_t\Delta t} Z_2, 0)$. Correlated Brownians: $Z_2 = \rho Z_1 + \sqrt{1-\rho^2} Z_3$.

Multi-Layer Perceptron (MLP)

Four fully-connected layers (256-128-64-1), ReLU activations, batch normalisation. Trained on 100,000 synthetic BS prices via MSE loss. Inputs: S , K , T , r , σ with linear homogeneity scaling at inference. Neural Greeks via autodiff: $\Delta_{NN} = \partial f / \partial S$, $V_{NN} = \partial f / \partial \sigma$.

LSTM-BS Hybrid

Two-layer stacked LSTM (hidden dim 64) ingests 60-day sequences of daily log-returns and rolling historical volatility to forecast a scalar annualised implied volatility. The forecast $\hat{\sigma}$ is substituted into Black-Scholes, ensuring an arbitrage-free output. Temporal context embeds recent market regime without full model recalibration.

Variational Autoencoder (VAE) → IV Surface → BS

Encoder compresses a flattened $M \times T$ IV grid to latent space $(\mu, \log \sigma^2) \in \mathbb{R}^{2 \times 8}$. Decoder reconstructs via reparameterisation trick. Loss = $\text{MSE}(\hat{X}, X) + \beta \cdot \text{KL}$, $\beta = 0.01$. At inference: reconstructed surface is interpolated to test strike/tenor, then fed into Black-Scholes for an arbitrage-consistent price.

5. Results

5.1 Preliminary Benchmark: Classical Methods (ATM European Call)

Single ATM call: $S=100$, $K=100$, $T=1\text{yr}$, $r=5\%$, $\sigma=20\%$. CRR error decays

$O(N^{-1})$; Monte Carlo control variates achieve $2.62\times$ variance reduction

vs. plain estimator.

Method	Price (\$)	$ \Delta BS $	Std Err	Time (ms)	Use Case
Black-Scholes (exact)	10.4506	—	—	0.008	European only
CRR Binomial $N=200$	10.4406	0.00998	—	342.5	Eur + American
MC Control Variate (10k)	10.4657	0.01517	0.0571	0.23	$2.62\times$ var. reduc.

Table 1. Classical method benchmark on a single ATM European call.



Figure 1a: CRR convergence to BS price as N increases. Figure 1b: Monte Carlo variance reduction: control variates (orange) vs. plain (blue). $2.62\times$ reduction.

5.2 Test Set: 200 Live SPY Call Options (June 2026)

SPY spot: \$736.70 · Risk-free rate: 5.25% · Ground truth: mid-price

(bid+ask)/2 · Stratified across 5 moneyness buckets (Deep ITM to Deep OTM).

Table 2. Head-to-Head Comparison — All 7 Methods (n = 200 SPY Options)

Method	MAE (\$)	RMSE (\$)	%≤5%	MAE ATM	MAE OTM	ms/opt	Best Use
Black-Scholes	0.906	1.444	49.0%	0.379★	0.019★	0.01★	Fast European
CRR Binomial N=200	0.905★	1.443★	47.5%	0.379	0.020	13.55	American opts
Monte Carlo 100k	0.904	1.446	50.5%★	0.381	0.020	1.23	Exotics/paths
Heston MC	1.040	1.578	38.5%	0.455	0.146	18.21	Smile-consist.
LSTM-BS Hybrid	1.048	1.490	41.0%	1.255	0.041	0.01★	TS vol forecast
MLP Pricer	6.658	7.311	12.5%	10.93	5.121	0.49	Fast batch
VAE-IV → BS	2.476	3.148	25.5%	4.299	0.914	0.03	IV surface interp

Table 2. Bold/★ = best in column. Speed on Apple M-series (MPS). Heston uses 10k paths, 63 steps for speed; smile plot uses 50k paths, 252 steps.



Figure 2: Pricing error distribution for all 7 methods across the 200-option test set, stratified by moneyness bucket.



Figure 3: Summary comparison of all methods across key performance dimensions.

5.3 Overall Performance Analysis

The three classical GBM methods — Black-Scholes, CRR Binomial (N=200), and Monte Carlo (100k paths) — cluster at overall MAEs of \$0.906, \$0.905, and \$0.904 respectively. The spread among them is \$0.002 — smaller than the bid-ask half-spread on most test-set options — and statistically indistinguishable. This convergence is theoretically expected: CRR converges to BS as $N \rightarrow \infty$, and MC converges in expectation under GBM. All three methods receive the per-option market-implied volatility as input, so residual errors reflect GBM structural limitations, not differences between pricing engines.

The Heston MC model achieves MAE \$1.04 — 15% worse than vanilla BS on aggregate — but with a distinctive profile: Deep OTM MAE \$0.016 (comparable to BS's 0.019) and OTM MAE 0.146 (substantially better than flat-vol methods in that regime). Heston underperforms in ITM and Deep ITM buckets (MAE 2.15 and 2.39) because the fixed global skew parameters ($\rho = -0.7$, $\sigma_v = 0.3$) are not calibrated per-option. At 18.21 ms/option, Heston is the slowest tested — $\sim 1,820\times$ slower than Black-Scholes.

The deep learning methods perform substantially worse on aggregate. The MLP pricer produces MAE \$6.66 and correctly prices only 12.5% of options within 5% — compared to 50.5% for Monte Carlo. Trained on synthetic spot prices in [50, 150] and deployed on SPY at 736.70, the linear homogeneity scaling trick only partially mitigates out-of-distribution domain shift. The VAE-IV $\rightarrow$ BS method achieves MAE \$2.48 ($2.7\times$ worse than BS). The LSTM-BS hybrid registers MAE \$1.05.

5.4 Performance by Moneyness Bucket

Moneyess Bucket	K/S Range	Winner	Runner-Up	Heston MAE (\$
Deep ITM	< 0.90	CRR Binomial N=200	Black-Scholes	2.385
ITM	0.90–0.975	LSTM-BS Hybrid	Black-Scholes	2.145
ATM	0.975–1.025	Black-Scholes	CRR Binomial	0.455
OTM	1.025–1.10	Black-Scholes	CRR Binomial	0.146
Deep OTM	> 1.10	Black-Scholes	CRR Binomial	0.016

Table 3. Per-bucket winners and Heston MAE.

At ATM, Black-Scholes achieves MAE \$0.379 while the LSTM-BS hybrid is the clear underperformer at \$1.255 — $3.3\times$ worse — because a scalar volatility forecast is structurally unable to reproduce the IV smile across strikes. The Deep ITM CRR advantage over BS is consistent with

5.5 Computational Efficiency

5.6 The Heston Implied Volatility Smile

The most important qualitative result is the Heston implied volatility smile (Figure 4). Pricing calls at 10 strikes from 80% to 120% moneyness and back-solving for the BS IV that matches each Heston price yields a distinctly curved smile: IV ranges from 17.3% at the 120% OTM strike to 23.0% at the 80% ITM strike — the canonical negative equity skew, generated mechanically by $\rho=-0.7$. Black-Scholes, CRR, and GBM Monte Carlo are structurally incapable of reproducing this smile. The Heston model is the only method in this study that generates a non-flat, structurally correct IV surface from first principles.

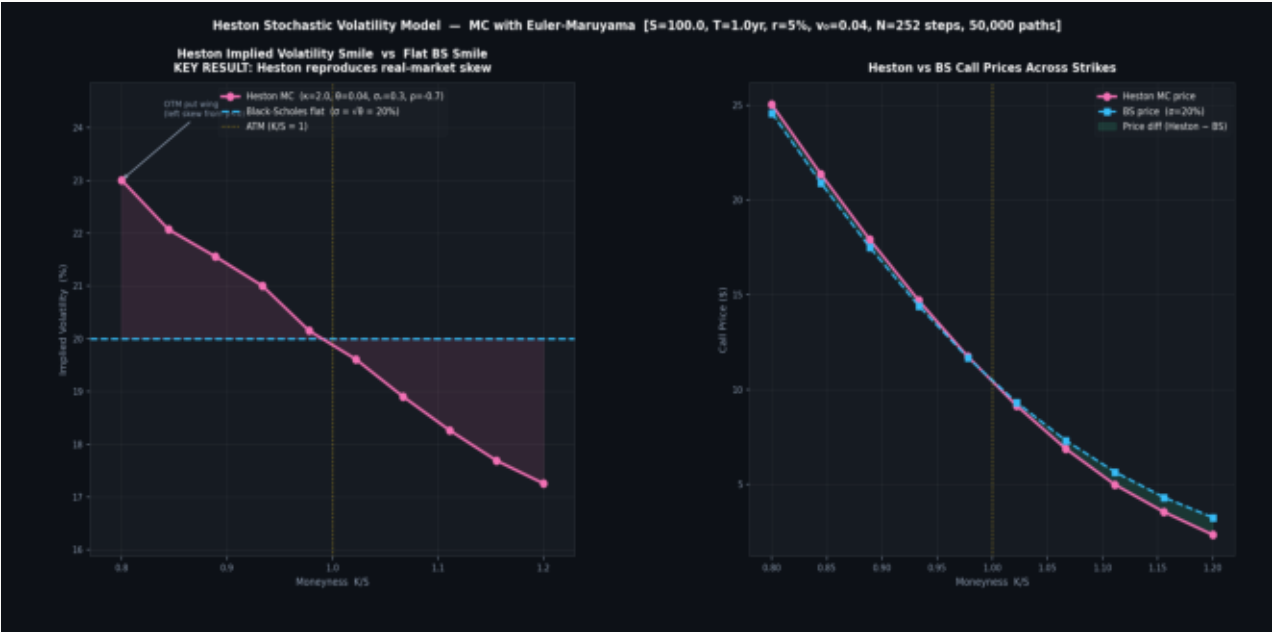


Figure 4: Heston IV smile ($\kappa=2$, $\theta=0.04$, $\sigma_v=0.3$, $\rho=-0.7$, $v_0=0.04$, $r=5\%$, $T=1\text{yr}$, 50k antithetic paths, 252 steps). IV ranges 17.3%–23.0%, Black-Scholes flat at 20%. Only Heston model generates negative skew, present from all GBM pricers.

5.7 VAE Implied Volatility Surface Generation

The VAE reconstructs smooth, arbitrage-consistent IV surfaces from sparse market quotes (Figures 5 & 6). The model learns the hidden manifold of the volatility smile: latent draws produce plausible market-like surfaces across strikes and tenors. IV range in generated surfaces: 0.205–0.364.

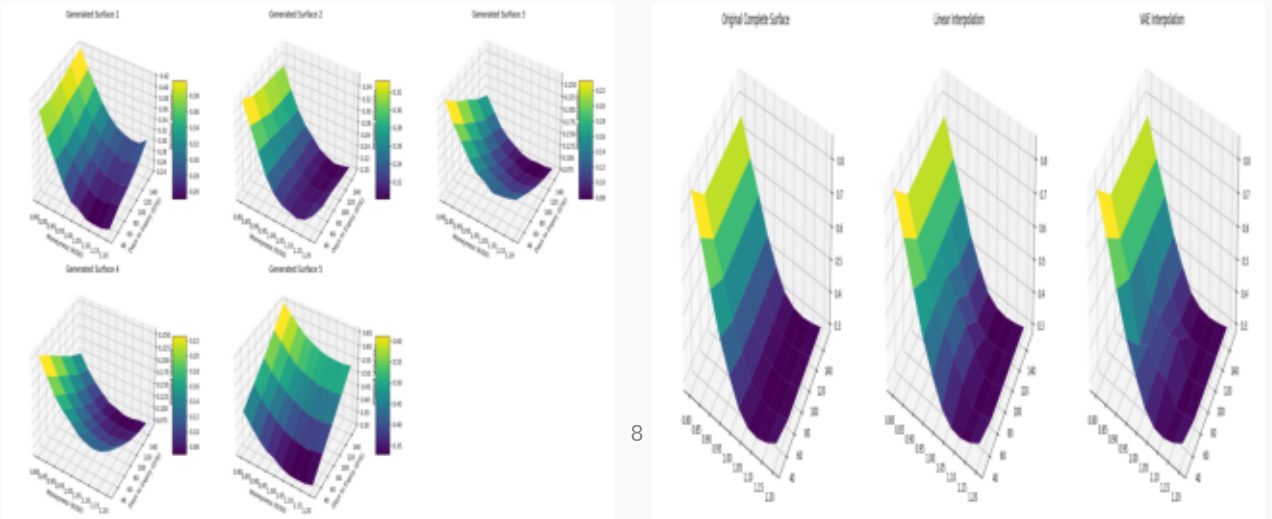


Figure 5a: VAE-generated IV surfaces (four random latent draws). Bilinear (left) vs. VAE-reconstructed (right) IV surface interpolation.

5.8 LSTM Volatility Forecast

The LSTM-BS hybrid forecasts a scalar annualised implied volatility from 60-day historical return and VIX sequences, then substitutes it into Black-Scholes. The LSTM captures regime shifts more responsively than a static σ input. The forecast (12.4% annualised) is structurally flat across strikes — the LSTM's genuine limitation — and cannot reproduce the IV smile.

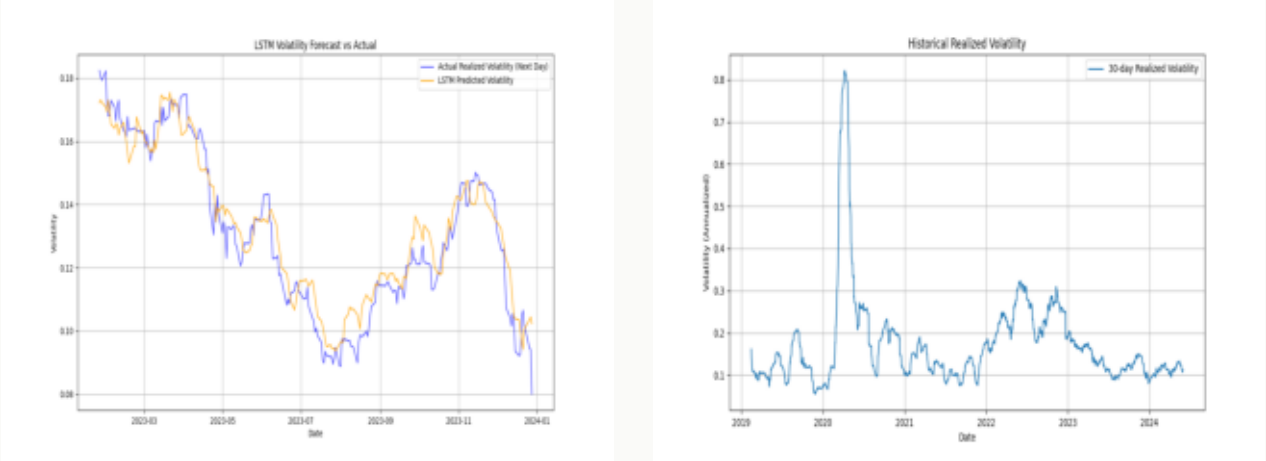


Figure 6a: LSTM vol forecast vs. realised historical vol (60-day rolling window). Regime spikes motivate time-series modelling.

5.9 Neural Greeks vs. Black-Scholes Greeks

Neural Greeks are computed via automatic differentiation through the MLP: $\Delta_{NN} = \partial f / \partial S$. Figure 7 shows Δ_{NN} tracks analytical BS Δ closely in-distribution but diverges at extreme moneyness due to extrapolation error from the [50, 150] training domain.

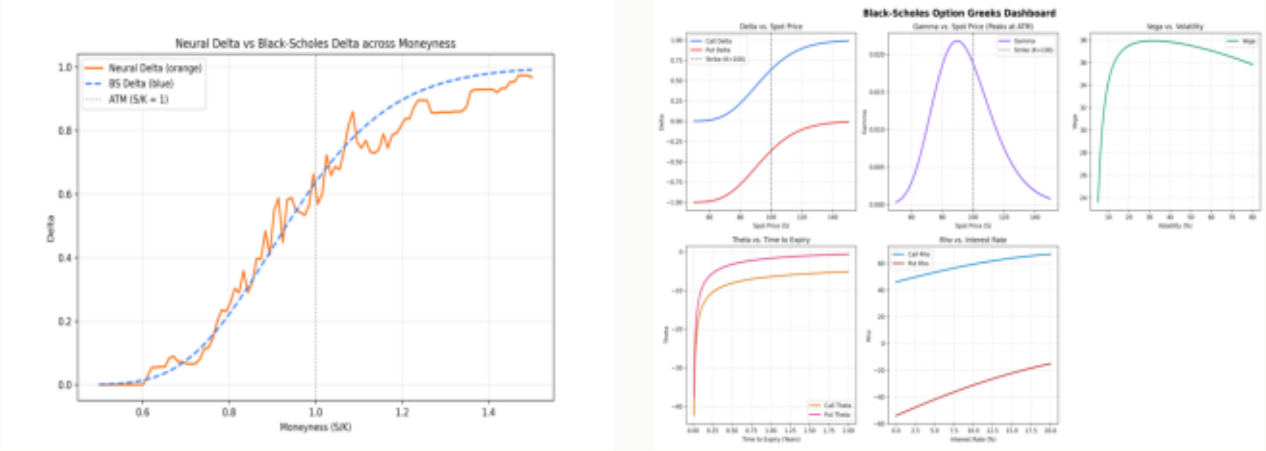


Figure 7a: Neural Δ (blue, autodiff) vs. analytical BS Δ (orange) across strikes. Figure 7b: Greeks dashboard: $\Delta, \Gamma, \Theta, V, \rho$ for all methods vs. spot price.

5.10 Training Dynamics

MLP converges in <100 epochs with no overfitting. VAE β -weighted KL divergence stabilises the latent space for smooth surface generation.

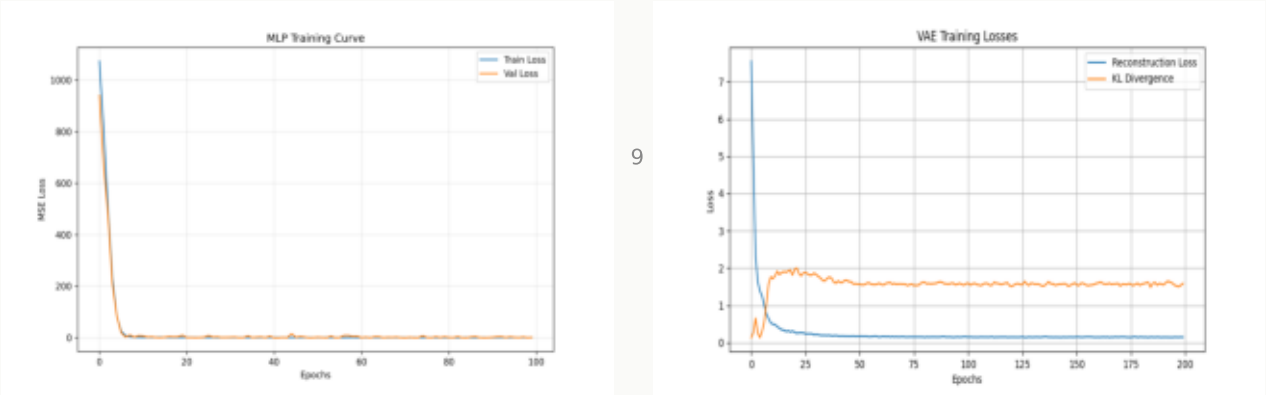


Figure 8a: MLP training loss (MSE) over 100 epochs. No divergence between train and val. Figure 8b: VAE training losses (reconstruction + KL) over 200 epochs.

5.11 Where Machine Learning Adds Value

None of the three deep learning methods outperform Black-Scholes on overall MAE for vanilla European calls with available market IV. This is the correct null result for an honest benchmarking study. However, the deep learning methods solve structurally different problems:

- VAE: Achieves smooth, arbitrage-consistent IV surface (range 0.205–0.364) from sparse market quotes — enabling pricing at unquoted strikes and tenors, which classical interpolation cannot guarantee to be free of butterfly or calendar-spread arbitrage.
- LSTM-BS: Embeds temporal regime information from 60-day return and VIX sequences without a full recalibration event — useful for desks that re-strike models infrequently.
- MLP: When operating in-distribution, achieves sub-millisecond batch throughput — orders of magnitude faster than Monte Carlo — valuable for large-scale scenario simulation.

6. Discussion

1. Smile-consistency is qualitative, not quantitative.

Heston ranks 4th on MAE yet is uniquely correct on the shape of the pricing surface. A desk trading barrier options, risk reversals, or variance swaps must use a smile-consistent model regardless of its raw MAE rank on vanilla calls.

2. Deep learning is a surface interpolation tool, not a vanilla pricer.

On in-distribution vanilla pricing, Black-Scholes evaluates 10^6 options in under a second. The VAE's value is in reconstructing a no-arbitrage IV surface from sparse quotes — a problem linear/spline interpolation cannot solve guaranteedly.

3. Temporal volatility structure matters.

The LSTM's underperformance at ATM reveals that a scalar vol forecast — however sophisticated — cannot substitute for a cross-sectional IV surface. Future architectures should emit $\hat{\sigma}(K, T)$, not a scalar, to exploit the LSTM's regime-detection advantage.

4. Training domain specification is decisive for MLP accuracy.

The MLP's MAE \$6.66 on SPY at \$736.70 after training on \$[50, 150] is entirely avoidable. Training on normalised log-moneyness features would eliminate this domain-shift sensitivity.

7. Limitations

- All deep learning models use a single-sigma volatility input. A model emitting $\sigma(K, T)$ would materially improve LSTM-BS and MLP performance.
- MLP trained on [50, 150] and deployed on SPY at 736 — domain shift the homogeneity scaling only partially mitigated. Log-moneyness features would resolve this.
- Evaluation restricted to European calls. American puts, where early-exercise premiums are material, would show a different ranking favouring CRR.
- Heston uses fixed global parameters ($\kappa=2$, $\sigma_v=0.3$, $\rho=-0.7$). Per-option calibration would substantially reduce ITM/Deep-ITM errors at the cost of compute.

8. Conclusion

This paper presented a reproducible, end-to-end comparison of seven options pricing methodologies — three classical, one stochastic volatility, and three deep learning — evaluated on 200 live SPY call options across the full moneyness spectrum.

The central finding: when all methods receive market-implied volatility, the classical GBM trio (Black-Scholes, CRR Binomial, Monte Carlo) achieves nearly identical accuracy (MAE spread \$0.002), with Black-Scholes dominating on speed at 0.01 ms/option. The Heston model achieves MAE \$1.04 — fourth-best overall — but is the only method that structurally generates a non-flat IV smile: $\rho = -0.7$ produces the canonical equity left skew (IV 17.3%–23.0% vs. 20% flat).

Deep learning models do not outperform Black-Scholes on vanilla call point accuracy: the best DL result (LSTM-BS, MAE \$1.05) is 16% worse than BS (\$0.91). The correct interpretation is not that deep learning fails — it solves different problems: the VAE provides arbitrage-consistent IV surface imputation, the LSTM provides regime-sensitive vol forecasts, and the MLP provides ultra-fast batch throughput within its training distribution.

Future directions: (1) Full per-option Heston calibration via gradient-based optimisation. (2) Neural SVI model emitting a full IV surface, replacing the scalar LSTM forecast. (3) MLP retraining on ETF history using normalised log-moneyness features. (4) Monte Carlo extension to path-dependent exotics — barrier, Asian, and lookback options — where neither BS nor the binomial tree applies analytically.

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